Banking on Persistence: The Impacts of a Credit Guarantee on Bank Lending in Indonesia

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Concerns about sub-optimal lending by banks

- Banks are important to economic prosperity.
  - Provision of liquidity, 'mobilisation of savings, evaluation of projects, management of risk, monitoring of managers, and facilitation of economic transactions' (King and Levine 1993; Schumpeter 1934; Croitoru 2012).

- Important function in allocating capital through lending.

- Concern about under-provision of capital by banks in emerging market countries, especially to key segments such as micro, small and medium enterprises (MSMEs) (Ayyagari et al, 2012).
  - In Indonesia there was a concern that banks were lending much less than capacity, and MSMEs were particularly out of the credit market.
  - Many other, more aggressive, programs had largely failed.

Perhaps credit guarantees can stimulate banks to lend more

- Some policymakers believe that banks need extra stimulus to enter new markets.
  - Accounts for experimentation costs, capacity investment costs, and extra risk from entering unfamiliar markets.

- A popular tool is the 'credit guarantee': a government or other donor provides banks with partial coverage on default risk.
  - E.g., if a firm takes a loan for $1000, and has a $600 remaining balance on the loan and then defaults, the credit guarantee fund pays the bank $300 under a 50% guarantee.
  - Leaves loan issuance in hands of bank, prices loans near market prices, and leaves some default risk with the banks, but still provides extra incentive to lend.

- Thousands of guarantees around the world, if anything increasing since the financial crisis in 2009 (Gozzi and Schmukler, 2016).

Do they generate more lending, and is it persistent?

- At least two important questions to ask, to think about market failures guarantees might solve, and cost-benefit analysis:

1. Do they generate (bank-side or firm-side) **additionality, or substitution**?
   - Credit guarantee schemes have seen tens of billions of dollars in government investment and hundreds of billions of loans issued around the world.
   - However these volumes do not prove that the programs led to more lending, or more credit access. Loans that would have already happened could just be relabeled.

2. Do they generate **increases in non-guaranteed lending**?
   - Guarantee might motivate the bank to build up internal loan screening and enforcement capacity (e.g., adopt credit scoring technology, start an SME lending unit, etc.). Might give bank staff useful experience in lending to certain sectors, improving their screening technology.
   - Might lead to spillovers to current MSME lending, and/or persistence.

What we do

- Develop a simple model to show how a credit guarantee can increase a bank's willingness to lend to a firm of a given quality, and extend the lending model to consider concurrent and dynamic increases in non-guaranteed lending.

- Test the predictions of the model with a differences-in-differences empirical strategy, using data from one of the largest credit guarantee programs in the world.

- Find that
  - KUR generates a significant increase in lending, and that this seems to represent real additionality (not significantly crowding out other lending, on net).
  - KUR spills over to other non-guaranteed lending.
  - Larger effects on sectors receiving little lending.

We provide one of the first studies addressing these questions

- Little rigorous empirical literature on the economic impacts of such schemes, especially in light of how widespread they are (Gozzi and Schmukler, 2016).

- Contributions:
  1. Add to literature showing that developing country banks may face barriers to lending, showing that a credit guarantee can help overcome this;
  2. Contributes a rigorous empirical evaluation to the global literature on credit guarantees, providing evidence for bank-side additionality and persistence;
  3. Results consistent with the idea that banks learn to screen borrowers through experience (informs literature on bank capacity building).

Key literature

- Banks value information
  - Informs their contract terms and lending capacity (Diamond 1984; Boyd and Prescott 1985; Ramakrishnan and Thakor 1984)
  - MSMEs are informationally opaque – huge asymmetries (Berger and Udell 2002, 2006; OECD n.d.). Banks are less skilled at collecting 'soft information' about them, can cause misallocation of credit.

- Relationship lending and bank learning
  - Banks learn about individual borrowers (Petersen and Rajan 1994; Berger and Udell 1995; Sharpe 1990)
  - Can banks also learn to overcome ex-ante screening problem when facing new borrowers?

- Some theoretical literature on guarantees (e.g., Doepke and Townsend 2001; Benavente et al 2006).

- Empirical studies large descriptive. Those with identification strategy use quasi-experimental designs to focus on firm-level additionality (e.g., Cowan et al 2012, Harigaya 2016).

Kredit Usaha Rakyat ("Peoples' Business Credit")

- Launched November, 2007. Active until late 2014, when program was suspended, then renewed and expanded from 2016 onward by Jokowi government.

- Three primary objectives of KUR: 1. poverty reduction; 2. job creation; 3. income generation.

- Structure
  - Government designs policy, licenses guarantors, convenes roundtables.
  - Two guarantors (Jamkrindo and Askrindo) – supervised by Bank of Indonesia.
  - Provides "mikro" (0-$2000) and "ritel" ($2000-$50,000) loans.
  - Effectively guarantees 70% (possibly 80%) of loans, at cost of about 2-3% of the value of the guarantee, which government pays for.

Kredit Usaha Rakyat ("Peoples' Business Credit")

- Originally 6 of the largest state-owned banks joined, and program was expanded to regional state-owned banks.

- Size of program: 14 trillion Rupiah in lending in 2008 (approximately $1.4 billion AUD); 100 trillion Rupiah in 2016 (approximately $10 billion AUD).
  - Government contribution about 3% of guarantee (70%), so annual cost 30 million AUD to 210 million AUD.

Setup of the model

- Model follows Marquez and Dell'Ariccia (2004).

- Entrepreneurs
  - Require investment $\boldsymbol{I}$ for proposed projects
  - Can have project type 'good' $\boldsymbol{G}$ or 'bad' $\boldsymbol{B}$, respectively (adverse selection)
  - If G, payoff $I(1+i)$

- Bank
  - Assume a single bank
  - Costless access to capital (lend from their own reserves)
  - Lends at a nominal rate $\boldsymbol{i}$
  - θ is the conditional probability a bank can correctly pick a good entrepreneur in a given market
  - Lends to satisfy zero expected profit condition (Jakab and Kumhof, 2014)

$$\mathrm{E}[\Pi] = \theta R - I \geq 0$$

Lending rules

- Rearranging the expected profit function gives us the following lending rule:

$$i \geq \frac{1}{\theta} - 1$$

- Lending to unfamiliar markets
  - Low θ → high interest rate
  - There is an interest rate threshold above which good borrowers, $\boldsymbol{G}$, do not borrow, i_G
  - i_U is interest rate when lending in an unfamiliar market (given θ)
  - If $i_U > i_G$ -- good borrowers do not borrow and banks do not lend

- Guarantee is represented by ω, where $0 \leq \omega \leq 1$

$$i_{CGS} \geq \frac{1}{(1-\omega)\theta+\omega} - 1$$

If, $i_{CGS} < i_G$ banks will lend.

Dynamic learning process

- Assume banks learn about screening borrowers through participation in the market
  - Allows them to update their 'lending technology.'

- Concave learning function such that:

$$\theta = \theta(t)$$

$$\frac{\partial \theta}{\partial t} > 0$$

$$\frac{\partial^2 \theta}{\partial t^2} < 0$$

- If the new probability of picking a good borrower is such that $i_N < i_G < i_U$ is satisfied – banks will continue to lend after the credit guarantee is removed.

Main predictions from the model

1. Additionality: banks should lend more on net while the credit guarantee is active (additionality).

2. Persistence: if the credit guarantee helps banks overcome lumpy initial learning costs, then increased lending will persist beyond scheme.

3. Impacts should be higher in markets with lower initial market penetration (i.e., more room for increases in θ).

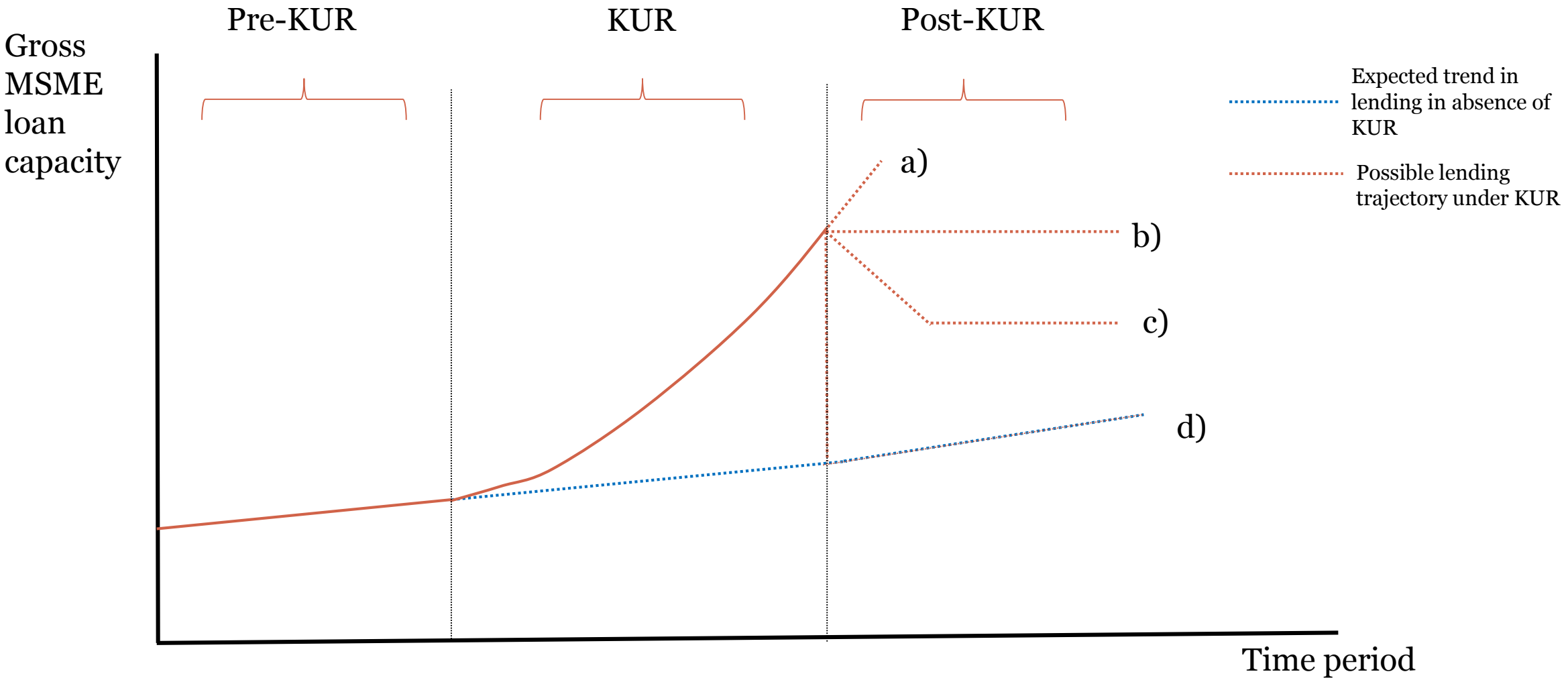
Empirical strategy: sources of variation

- Two sources of variation:

1. The KUR was limited to a relatively small set of state-owned banks (6 major banks, later (much smaller) regional banks).
   - Idea is that non-KUR banks' MSME lending provides a counterfactual for the *trend* in MSME lending of KUR banks. This allows us to "difference out" MSME lending that banks would have already done without KUR.

2. The KUR was temporarily suspended at the end of 2014 (after Jokowi government elected in July, 2014), with significant uncertainty about whether the program, or which banks, would continue afterward.
   - This allows for a short-run test of whether MSME lending persisted after the KUR suspension.

Idea of the empirical strategy: during and after KUR



Data: SME lending by banks in Indonesia: 2005-2016

1. Condensed Financial Statements from all Indonesian banks curated by Indonesian financial services authority (OJK);

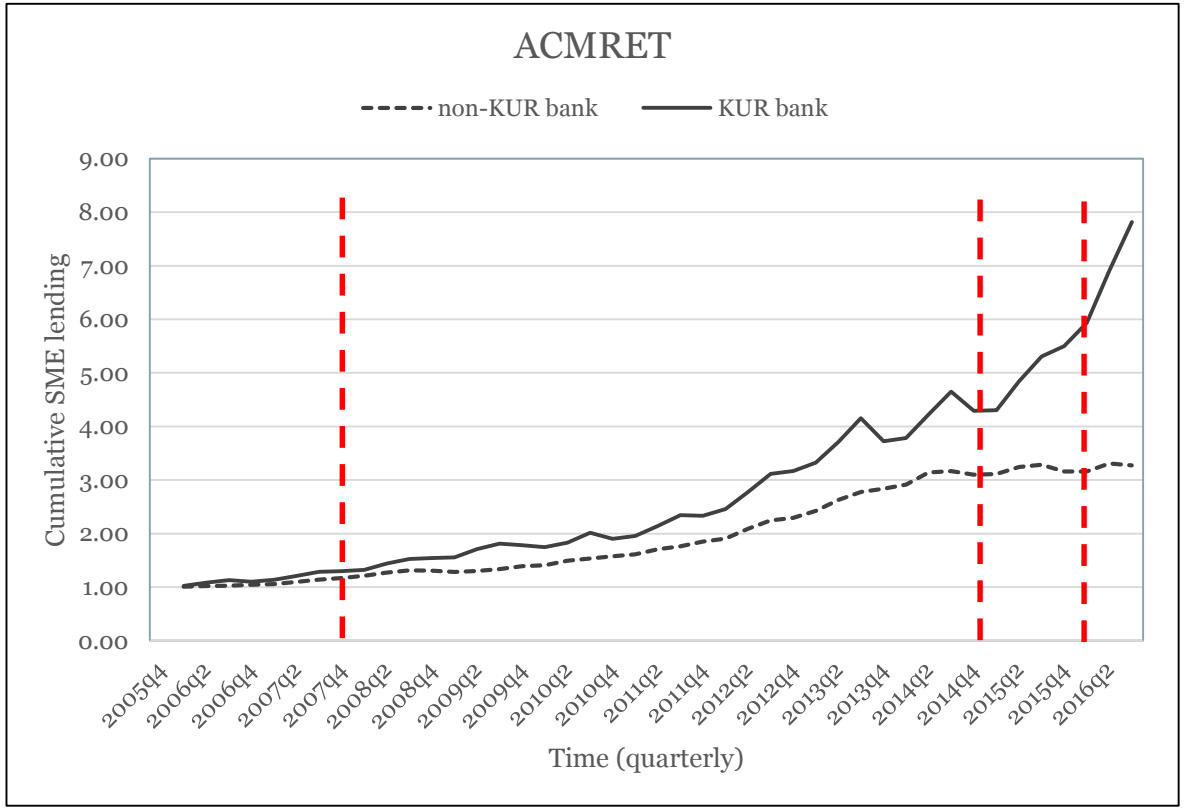
2. Indonesian Banking Statistics (SPI) published by the central bank of Indonesia - Bank Indonesia (BI) - and OJK;

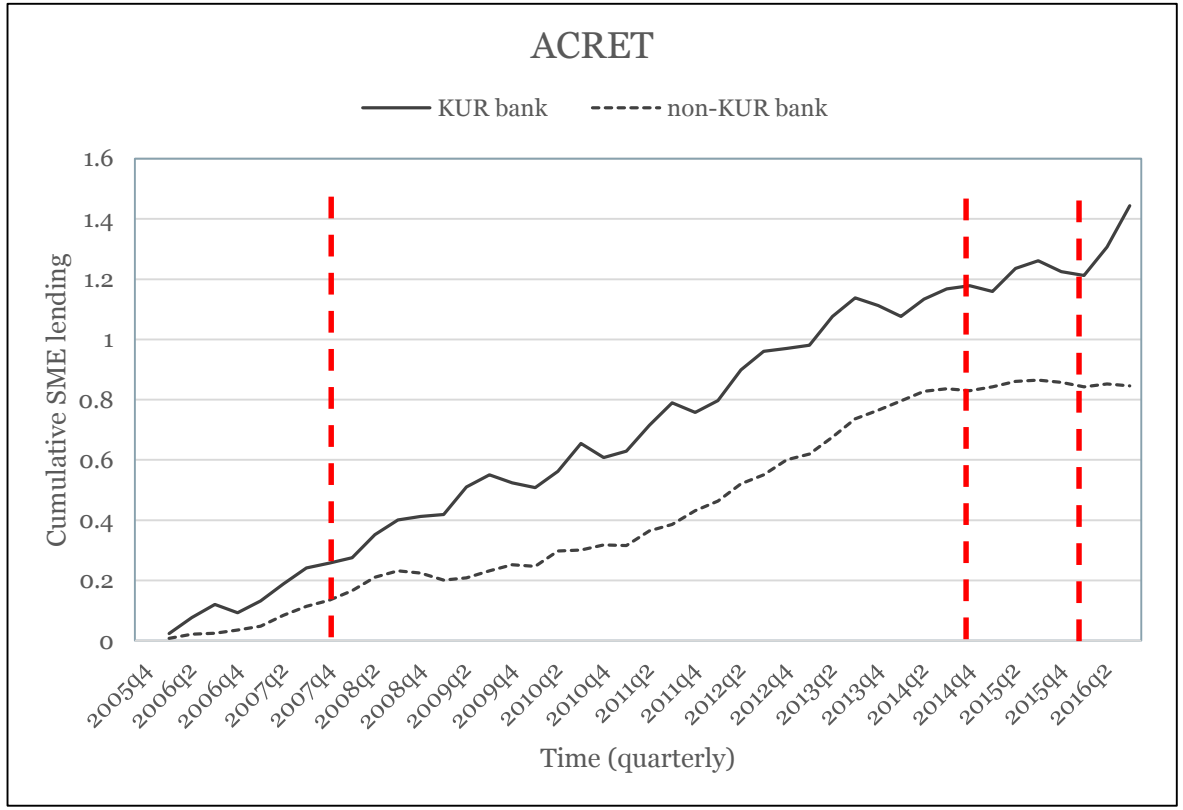
3. KUR lending guarantee data from guarantee firms (Jamkrindo and Askrindo); and

4. Total KUR disbursement data reported by the public body that oversees the program, the Coordinating Ministry for Economic Affairs of the Republic of Indonesia.

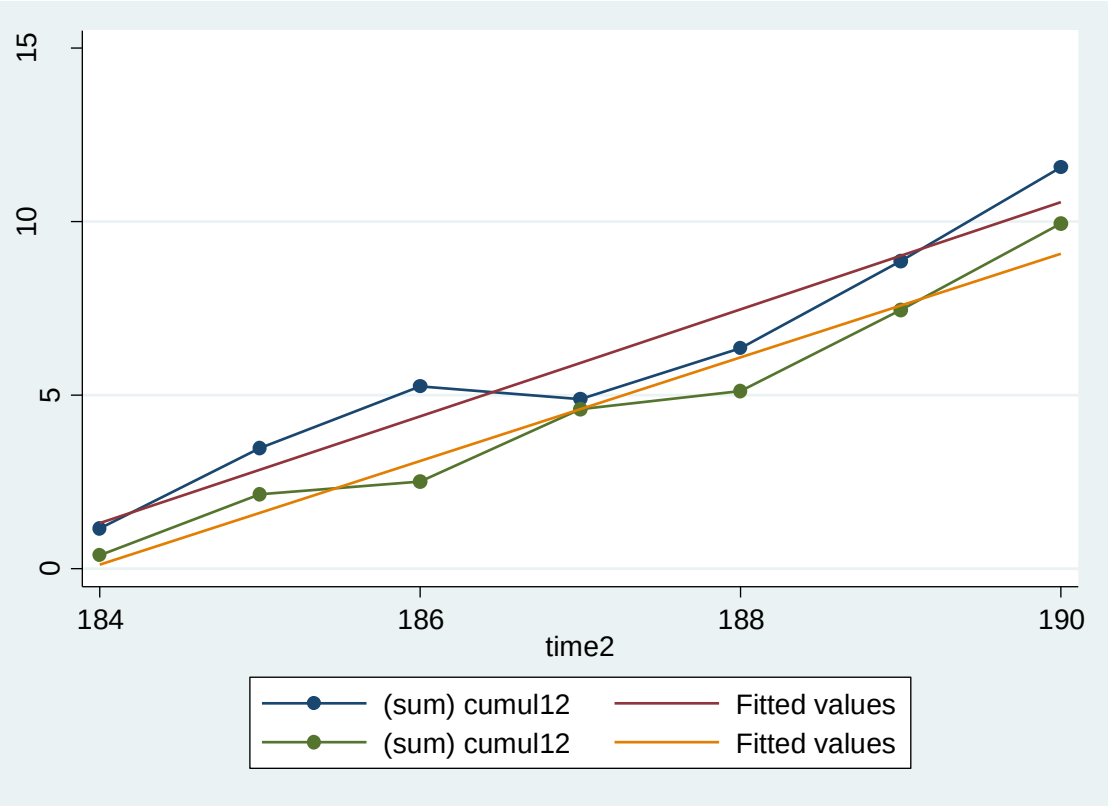
Descriptive evidence on lending dynamics

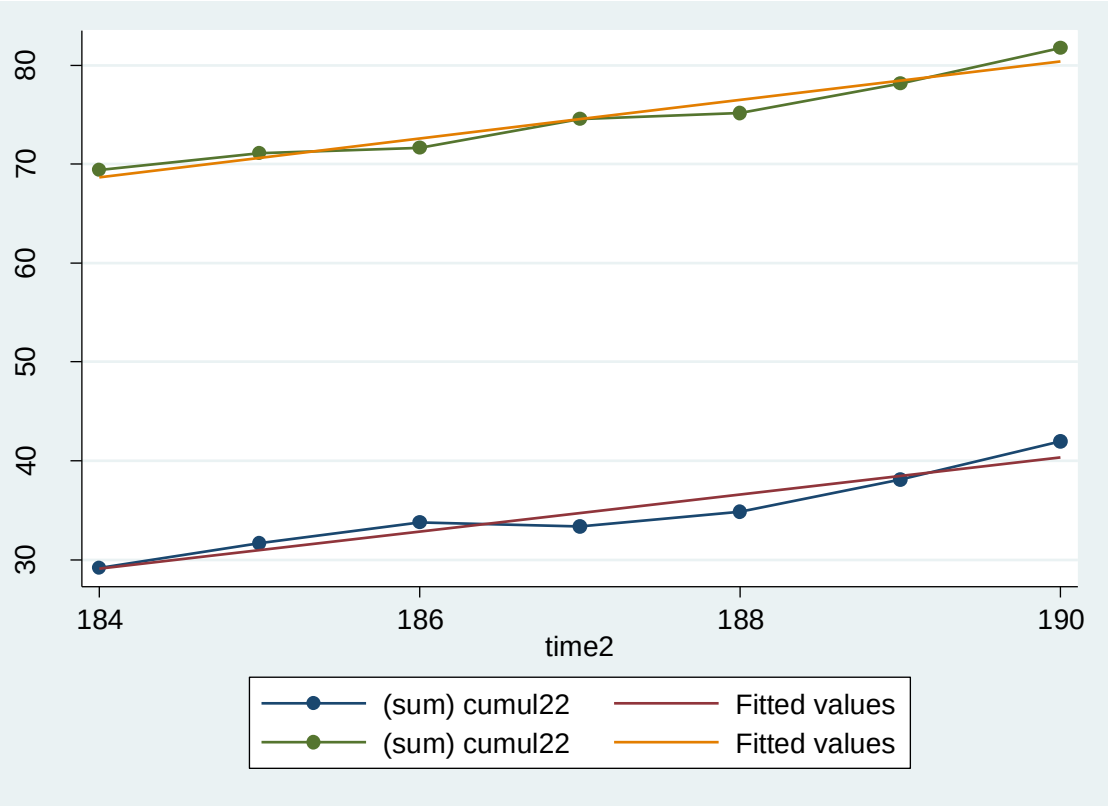
Follow Lee and Mas (2012) in calculating cumulative and average weighted cumulative returns, also suggestive of common pre-trends.





Validation of parallel pre-trends (pre-Q4 2007)





Also report on regression-based test which finds no significant difference in linear pre-trends in lending between KUR and non-KUR banks

Testing for additionality and persistence

$$CumulativeLending_{tb} = \beta_0 + \beta_1 KURperiod + \beta_2 KURsuspension + \beta_3 KURbanks + \beta_4 KURperiod \times KURbanks + \beta_5 KURsuspension \times KURbanks + X_{tb}\beta' + \varepsilon_{tb}$$

- $CumulativeLending_{tb}$: increase in lending for bank b in period t.
- *KURperiod* and *KURsuspension* are dummies for 2008-Q42014 and 2015 onward, respectively.
- *KURbanks* identifies the KUR banks.

Testing for additionality and persistence

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- Predictions:
  - (prediction 1) If bank-side additionality, then $\beta_4 > 0$.
  - (prediction 2) If strong persistence, then $\beta_5 \sim \beta_4$.
  - (prediction 2) If persistence, then $\beta_5 > 0$.

Testing for additionality and persistence

	POLS	RE	Bank Fixed Effects		
	(1)	(2)	(3)	(4)	(5)
	MCRET	MCRET	MCRET	MCRET	MCRET
KURsuspension	-0.855**	-0.856***	-0.856***	-5.483	1.660
	(0.35)	(0.27)	(0.27)	(12.10)	(14.38)
KURbank	0.159	0.159	0.000	0.000	0.000
	(0.24)	(0.46)	(.)	(.)	(.)
KURperiod	-0.103	-0.103	-0.103	-0.105	-0.105
	(0.26)	(0.20)	(0.20)	(0.20)	(0.20)
KURperiod*KURbank	1.122***	1.118***	1.118***	1.118***	1.118***
	(0.27)	(0.20)	(0.20)	(0.20)	(0.20)
KURsuspension*KURbank	2.454***	2.444***	2.444***	2.443***	-21.917
	(0.51)	(0.39)	(0.39)	(0.39)	(26.53)
Intercept	1.130***	1.130***	1.174***	1.171***	1.171***
	(0.20)	(0.27)	(0.14)	(0.14)	(0.14)
Time trends	YES	YES	YES	YES	YES
KUR period-specific trends	NO	NO	NO	YES	YES
Macroeconomic controls	NO	NO	NO	NO	NO
Time Fixed effects	NO	NO	NO	NO	NO
Hausman test (2) and (3)	0.03				
R^2	0.145	0.1942	0.194	0.194	0.194
N	3671	3671	3671	3671	3671

Testing for additionality and persistence

- Results suggest that:
  - The KUR generates additionality: 112% increase above non-KUR banks.
  - KUR generates persistence, though issues in disentangling from time trends.
- Additional robustness checks:
  - Non-linear, period-specific, time trends.

Testing for persistence in sectors with less lending

- In 2012 KUR program identified 4 priority sectors (agriculture, forestry, light manufacturing) to get an 80% guarantee.

$$logUMKMlending_{ts} = \beta_0 + \beta_1 PriorityPeriod_t + \beta_2 PrioritySectors + \beta_3 Prioritysector \times PriorityPeriod_{ts} + \beta_4 KURsuspension_t + \beta_5 KURsuspension \times Prioritysector_{ts} + \varepsilon_{ts}$$

- $PriorityPeriod_t$: priority sector identifier.

- $PriorityPeriod_{ts}$: post-2012 priority period.

Testing for persistence in sectors with less lending

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$$logUMKMlending_{ts} = \beta_0 + \beta_1 PriorityPeriod_t + \beta_2 PrioritySectors + \beta_3 Prioritysector \times PriorityPeriod_{ts} + \beta_4 KURsuspension_t + \beta_5 KURsuspension \times Prioritysector_{ts} + \varepsilon_{ts}$$

- (prediction 3) $\beta_5 > 0$ if see greater persistence in sectors with less access to finance ex ante (e.g., priority sectors).

Testing for greater persistence in priority sectors

	OLS	RE	Sector Fixed Effects			
	(1) logumkm	(2) logumkm	(3) logumkm	(4) logumkm	(5) logumkm	(6) logumkm
Priorityperiod	0.209	0.209***	-0.128*	0.209***	-0.128*	-0.119
	(0.17)	(0.03)	(0.07)	(0.03)	(0.07)	(0.14)
Priority sector	0.146	0.146	0.146	0.000	0.000	0.000
	(0.42)	(1.46)	(1.46)	(.)	(.)	(.)
Priority sector*Priority Period	0.328	0.328***	0.328***	0.328***	0.328***	0.328***
	(0.49)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)
KUR suspension	0.272	0.272***	-0.185**	0.272***	-0.185**	-0.159
	(0.25)	(0.05)	(0.08)	(0.05)	(0.08)	(0.19)
KUR suspension*Priority Sector	0.517	0.517***	0.517***	0.517***	0.517***	0.517***
	(0.71)	(0.15)	(0.14)	(0.15)	(0.14)	(0.15)
Quarter fixed effects	NO	NO	NO	NO	NO	YES
Time trends	NO	NO	YES	YES	YES	YES
R^2	0.009	0.0895	0.1401	0.090	0.140	0.154
N	901.000	901.000	901.000	901.000	901.000	901.000

Testing for greater persistence in priority sectors

- Results suggest that:
  - The KUR generates stronger additionality in the priority sectors, consistent with idea that we see higher returns where banks have more to learn.
  - However need to be cautious since we are not distinguishing KUR and non-KUR banks.
- Additional robustness checks:
  - Non-linear, period-specific, time trends.

Intensive margin effects

- Use bank-level KUR quotas as instruments to provide some exogenous variation in *levels* of KUR lending, to study the effects of KUR expansion on the overall SME lending market in Indonesia.
  - "How much" does KUR stimulate the overall SME lending market (KUR and non-KUR banks)?

- Find that a 1% increase in KUR lending leads to 0.5% increase in overall SME lending.

Caveats and open questions

- What did banks believe about continuation of the program in 2015?
  - Outside indications of significant uncertainty; not clear what President Jokowi government's goals would be.
  - Set of banks significantly changed.
- KUR committee: what was role of government pressure?
  - But banks regularly overshoot targets, especially largest ones.
- We don't pin down mechanism for dynamic effects – what exactly is going on inside the banks?
- Is this actually good (or not) (for firms, households, etc)?
  - Harigaya (2017) shows that the program increased access to credit for micro and small enterprises, using variation in locations of the largest bank (BRI).

Conclusion

- Our findings suggest that:
  - KUR led to a net increase in lending (additionality); cumulatively about 350% over the course of the program.
  - It did not significantly crowd out counterfactual lending by KUR banks.
  - KUR led to spillovers on non-guaranteed lending in KUR banks.
  - Incentive changes in program led to increases in lending (e.g., increased guarantee for target sectors).

- Suggests KUR was a cost-effective instrument to increase MSME lending in Indonesia.

- Next steps:
  - Extending time period (backward—parallel trends-- and forward—better tests of persistence)

Thank you!

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